II Instanton Floer Homology

Lecture 11 // Sept 29 // Insanton Floer Homology

The rest of the course will follow this general outline.

1) broad overview of Insanton Floer Homology
2) formal properties & applications
3) more detailed disussion of (1)
4) ?? more recent papers

Instanton Floer Homology is a 3-manifold invariant. Originally, it was defined only for integer homology spheres, and we'll restrict to Y a $\mathbb{Z}HS^3$ for simplicity, but the constructions we'll discuss can be extended.

The general procedure is as follows:

$$Y \rightsquigarrow C(Y) \rightsquigarrow H_*(C(Y))$$

$\mathbb{Z}HS^3$ — Instanton chain complex — Instanton Floer homology

Where does $C(Y)$ come from? Two slogans:

1) ∞-dimensional Morse homology assc. to the CS functional

ie, there is a mysterious S^1-valued functional on an auxilliary space assc. to Y (like $\mathcal{A}/\mathcal{G}$, but for Y^3)

$$CS: \mathcal{A}/\mathcal{G} \longrightarrow \mathbb{R}/\mathbb{Z}$$

Whenever you have a $\mathbb{R}$-valued function, we take the morse hom:

- the cp's of CS become the generators of the chain complex
- the differentials are defined (roughly) by studying the gradient flow of CS.

2) The generators of $C(Y)$ are in correspondence with conjugacy classes of $SU(2)$ representations of $\pi_1(Y)$. So $C(Y)$ actually comes from π_1.

Today, we'll broadly discuss 1&2, and why they're the same.

What is the Chern-Simons Functional

Take Y^3 a $\mathbb{Z}HS^3$ (tho nothing today depends on that)

Fix a $SU(2)$ bundle over Y. Since $c_2(4)=0$, there's only one such bundle, the trivial one $E = Y \times \mathbb{C}^2 \to Y$

Just like before, we study $\mathcal{A} = \{ SU(2) \text{ con on } E^3 \}$

and $\mathcal{G} \curvearrowright \mathcal{A}$, $\mathcal{G} = \Gamma(\mathrm{Aut}(E))$. Since E is trivial, $\mathcal{G} \cong \mathrm{Maps}(Y, SU(2))$

Note that $\pi_0(\mathcal{G}) \cong \mathbb{Z}$ since $SU(2) \cong S^3$, and maps $Y \to S^3$ are determined by the degree $\deg(g) \in \mathbb{Z}$

$\mathcal{A}/\mathcal{G}$ = configuration space / gauge, the domain of CS

Def The Chern-Simons Functional is a map

$$CS : \mathcal{A}/\mathcal{G} \to S^1$$

$$CS(A) = \frac{1}{8\pi^2} \int_Y \mathrm{tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right)$$

By this, we mean first: fix a global trivialization of E.

In this global trivialization, ∇ is represented by a matrix of 1-forms A, which is the input of the integral.

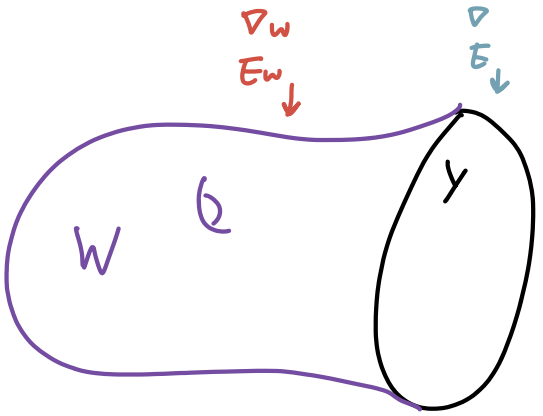
Exercise If $g \in \mathcal{G}$ is a global gauge transformation, $CS(g^* A) = CS(A) + \deg(g)$

This gives another way of saying that $\pi_0(\mathcal{G}) \cong \mathbb{Z}$; each connected component corresponds to a trivialization, so this is saying that, up to homotopy, there are $\mathbb{Z}$-many trivializations of E. The value of CS depends on which trivialization you choose.

This def is super explicit, but isn't as insightful as the following alt. def.

Def Let A be a connection on E. Choose any:

- smooth 4-mfd W^4 such that $\partial W \cong Y$
- $SU(2)$-bundle E_W on W such that $E_W \cong E$
- $SU(2)$-connection ∇_W such that $\nabla_W|_Y = \nabla$



Then $$CS(\nabla) = \frac{1}{8\pi^2}\int_W (F_{\nabla_W} \wedge F_{\nabla_W})$$

This definition follows from the definition of relative characteristic classes.

But as-is, you should be skeptical that this is well defined.

Let's choose a different W', $E_{W'}$, ∇'. Then

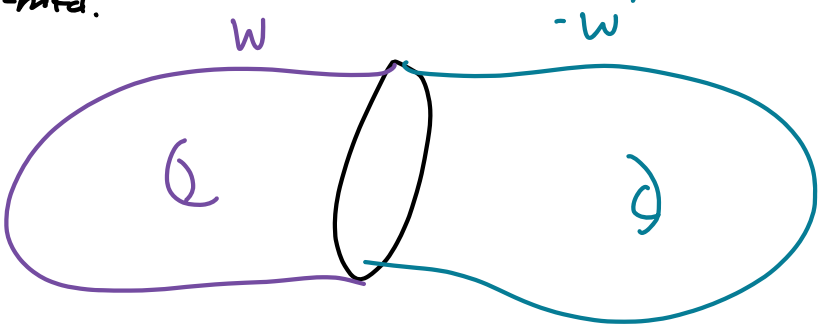
$$CS(\nabla) = \frac{1}{8\pi^2}\int_W (F_{\nabla_W} \wedge F_{\nabla_W}) \text{ and}$$

$$CS(\nabla') = \frac{1}{8\pi^2}\int (F_{\nabla_{W'}} \wedge F_{\nabla_{W'}})$$

If this definition is well-defined, then $CS(\nabla) - CS(\nabla') = 0$.

To compute the difference, glue W and $-W'$ along Y to get a $SU(2)$ bundle over a closed 4-mfd.

Then their difference



$$\frac{1}{8\pi^2}\int_W (F_{\nabla_W} \wedge F_{\nabla_W}) - \frac{1}{8\pi^2}\int_W (F_{\nabla_{W'}} \wedge F_{\nabla_{W'}})$$

$$= \left\langle c_2(E_\Sigma), [W \underset{Y}{\cup} -W'] \right\rangle \in \mathbb{Z}$$

Thus different W's change $CS(\nabla)$ by an integer.

Thus $CS(\nabla)$ is well-defined in S^1.

Remark The bundles E_W, $E_{W'}$ over W and W' glue together to a a well-defined bundle E_Σ over $W \cup_Y -W'$. The same is true for ∇_W and $\nabla_{W'}$, obtaining ∇_Σ

The Chern-Simons integral of this connection is that of a closed 4-manifold.

$$CS(\nabla_\Sigma) = \frac{1}{8\pi^2}\int_{W\cup_Y -W'} tr\left(F_{\nabla_\Sigma} \wedge F_{\nabla_\Sigma}\right) \in \mathbb{Z}$$

and splits as $\int_W tr\left(F_{\nabla_\Sigma|_W} \wedge F_{\nabla_\Sigma|_W}\right) - \int_{W'} tr\left(F_{\nabla_\Sigma|_{W'}} \wedge F_{\nabla_\Sigma|_{W'}}\right)$.

However, for the connections ∇_W and $\nabla_{W'}$ over 4-manifolds with non-empty boundary, $CS(\nabla_W) \notin \mathbb{Z}$.

In our proof of well-definedness, we relied on some gluing procedure without verifying what that procedure actually is: this was incredibly sloppy. We should also ask whether this def. plays well with G_g.

In order to glue E_W & $E_{W'}$ together, we must fix identifications $E_W|_Y \cong E_{W'}|_Y$. Since there is only one $SU(2)$ bundle over Y, the trivial one, we could argue that E_W, $E_{W'}$ are trivial bundles, thus E_Σ is trivial, so the difference is zero. While the first part of this is true, like we saw with $\mathbb{CP}^1$, two trivial bundles could be glued via some nontrivial boundary maps, like $\mathbb{Z}^n$, which results in a nontrivial bundle. Even simpler, two trivial $\mathbb{R}$-bundles over the interval I could be glued via a twist, resulting in the Möbius band.

The number of these different maps is the number of retrivializations of the bundle on the boundary. So we must ask: How many retrivializations of ∂W are there? The answer is $\mathbb{Z}$, as we saw earlier, and these $\mathbb{Z}$-many gluings give $\mathbb{Z}$-many $SU(2)$ bundles over W!

Claim Re-trivializing a bundle E_W over $\partial W \cong Y$ by composing with some $g \in G_J$ changes $c_2(E_\Sigma)$ by $\deg g \in \mathbb{Z}$.

Lemma The two definitions agree

Pf This follows immediately from Stokes' theorem: $\left(\int_W d\beta = \int_{\partial W} \beta \right)$

$$8\pi^2 \, CS(A) = \int_Y \operatorname{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A) \right)$$

$$= \int_W d\left(\operatorname{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A) \right) \right)$$

Simplifying, $d\left(\operatorname{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A) \right) \right)$

$$= \operatorname{tr}\left(d\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A) \right) \right)$$

$$= \operatorname{tr}\left(d(A \wedge dA) + \frac{2}{3} d(A \wedge A \wedge A) \right)$$

$$= \operatorname{tr}\left(dA \wedge dA - \underbrace{A \wedge d^2 A}_{0} + \frac{2}{3}\left(dA \wedge A \wedge A - A \wedge dA \wedge A + A \wedge A \wedge dA \right) \right)$$

$$= \operatorname{tr}\left(dA \wedge dA + \frac{2}{3} dA \wedge A \wedge A - \frac{2}{3} A \wedge dA \wedge A + \frac{2}{3} A \wedge A \wedge dA \right)$$

$$= \operatorname{tr}\left(dA \wedge dA + 2\, dA \wedge A \wedge A \right)$$

$$= \operatorname{tr}\left(dA \wedge dA + dA \wedge A \wedge A + A \wedge A \wedge dA \right)$$

$$= \operatorname{tr}\left((dA + A \wedge A) \wedge (dA + A \wedge A) \right) + \underbrace{\operatorname{tr}(A \wedge A \wedge A \wedge A)}_{0}$$

$$= \operatorname{tr}(F_A \wedge F_A)$$

$$\Rightarrow \int_Y \operatorname{tr}\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A) \right) = \int_W \operatorname{tr}(F_A \wedge F_A)$$

$$= 8\pi^2 \left\langle c_2(E_\Sigma), [W \underset{Y}{\cup} -W'] \right\rangle \quad \square$$

Rmk Timeline: this all happened pre donaldson, post relative characteristic classes.

Just like in Donaldson's theorem, we want to study the critical points of CS, which we'll denote by $crit(cs) \subset \mathcal{A}/\mathcal{G}$ What about interpretation #2?

Thm $\nabla \in \mathcal{A}/\mathcal{G}$ is a critical point of the Chern-Simons functional IFF ∇ is flat, ie, $F_A = 0$

Pf $\nabla \in Crit(CS)$

$$\Leftrightarrow \frac{d}{dt}\Big|_{t=0} CS(\nabla + tb) = 0$$

$$\Leftrightarrow \frac{d}{dt}\Big|_{t=0} 8\pi^2\, CS(\nabla + tb) = 0$$

$$\Leftrightarrow \int_Y tr\left((A+tb)\wedge d(A+tB) + \frac{2}{3}\left((A+tb)\wedge(A+tb)\wedge(A+tb)\right)\right) = 0$$

$$d(A+tB) = dA \wedge B - A \wedge dB$$

$$\Leftrightarrow \int_Y tr\left(B\wedge dA + A\wedge dB + \frac{2}{3} B\wedge A\wedge A + \frac{2}{3} A\wedge B\wedge A + \frac{2}{3} A\wedge A\wedge B\right) = 0$$

$$\Leftrightarrow \int_Y tr\left(2B\wedge dA + 2B\wedge A\wedge A\right) = 0$$

$$= 2\int_Y tr\left(B\wedge F_A\right) = 0 \qquad \forall B$$

$$\Leftrightarrow F_A = 0$$

$$grad_A\, CS = *F_A \qquad \square$$

Thm Critical Points of the Chern-Simons functional modulo gauge are in bijection with $SU(2)$ representations of $\pi_1(Y)$ modulo conjugation.

Def The Character Variety of a 3-mfd is

$$\mathcal{R}(Y) = \left\{ \begin{array}{l} SU(2) \text{ representations} \\ \text{of } \pi_1(Y) \end{array} \right\} / \sim$$

where $\rho_0 \sim \rho_1$ IFF $\rho_1 = g^{-1} \cdot \rho_0 \cdot g$

In this language, $\mathrm{Crit}(CS) \leftrightarrow \mathcal{R}(Y)$. We now prove the theorem.

Proof Taking the holonomy gives a map

$$\{\text{Flat connections}\} \to \left\{ \begin{array}{l} SU(2) \text{ representations} \\ \text{of } \pi_1(Y) \end{array} \right\}$$

Since $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$, this holonomy decends to a map on the quotient

$$\{\text{Flat connections}\}/\mathcal{G} \to \left\{ \begin{array}{l} SU(2) \text{ representations} \\ \text{of } \pi_1(Y) \end{array} \right\} / \sim \; = \mathcal{R}$$

It's not obvious this map is surjective.

To show surjectivity, let $\rho : \pi_1(Y) \to SU(2)$ be a rep^n of $\pi_1(Y)$. Construct an $SU(2)$ bundle E and connection ∇ on E as follows:

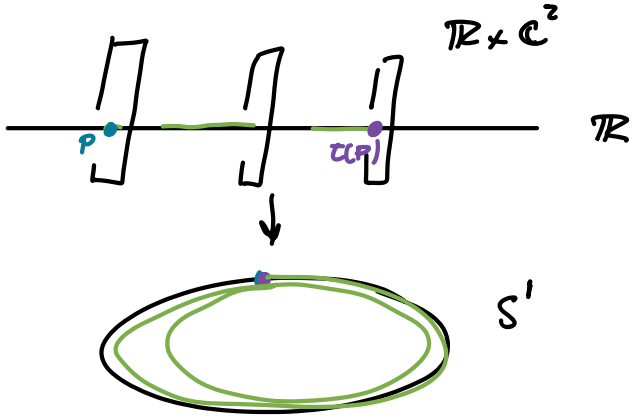
1) Form $\tilde{E} = \tilde{Y} \times \mathbb{C}^2$ trivial bundle over universal cover of Y

2) Quotient out $\tilde{E}$ by identifying the fibres $p \times \mathbb{C}^2$ and $\tau(p) \times \mathbb{C}^2$

for $\tau \in \mathrm{Deck}(\tilde{Y}) \cong \pi_1(Y)$

$\uparrow$ $\gamma \in \pi_1(Y) \leftrightarrow \tilde{\gamma}$ btwn $p, \tau(p)$

via the matrix given by $\rho(\gamma) \in SU(2)$, where γ lifts to $\tilde{\gamma}$



This defines a bundle E over Y, as well as parallel transport on E by pushing down parallel transport on $\widetilde{E}$

By construction, this parallel transport will have holonomy ρ.

This shows surjectivity.